ORF309 F2025 Midterm2

November 2025

Problem 1 (Insurance)**.** When customers file claims, the insurer's payment depends not only on the loss size but also on a variety of unpredictable factors such as regional regulations, claim-processing backlog, and negotiated discounts with repair shops and medical centers. As a result, the fraction of each loss that the policyholder (the insured) must pay out-of-pocket varies randomly from claim to claim. Assume:

- X = total loss amount (in thousands of dollars) with tail distribution

$$\mathbb{P}(X > x) = \begin{cases} 1, & x < 1, \\ \dfrac{1}{x^2}, & x \geq 1, \end{cases}$$

- U = out-of-pocket fraction, uniformly distributed in $[0, 1]$, independent of X.

The insured individual pays $Y = UX$ (in thousands of dollars). For example, if the insured individual gets a medical bill with total cost $X = 10$ thousand dollars, and the out-of-pocket fraction turns out to be $U = 0.1$, then the individual pays $Y = UX = 0.1 \times 10 = 1$ thousand dollars.

(a) On average, how much does the insured individual expect to pay?

(b) Determine the joint probability density function of (X, Y). (Hint: you may want to start with the conditional density of Y given X. Be careful of the range.)

(c) Suppose that after an incident the insured individual pays exactly 1 thousand dollars out-of-pocket. What is the probability that the total loss **exceeds** 2 thousand dollars?

Solution. (a) From the tail distribution of X, we can calculate its expectation as

$$\mathbb{E}[X] = \int_0^\infty \mathbb{P}[X > x]\,\mathrm{d}x = \int_0^1 1\,\mathrm{d}x + \int_1^\infty \frac{1}{x^2}\,\mathrm{d}x = 2.$$

As U and X are independent,

$$\mathbb{E}[Y] = \mathbb{E}[UX] = \mathbb{E}[U]\mathbb{E}[X] = \frac{1}{2} \times 2 = 1.$$

(b) Given $X = x$, Y is conditionally distributed as $\mathrm{Uniform}([0, x])$, which means the conditional density of Y given X is

$$p_{Y|X}(y|x) = \frac{1}{x}\mathbf{1}_{\{0 \leq y \leq x\}}.$$

The marginal density of X can be calculated by

$$p_X(x) = -\frac{\mathrm{d}}{\mathrm{d}x}\mathbb{P}[X > x] = \frac{2}{x^3}\mathbf{1}_{\{x\geq 1\}}.$$

As a result, the joint density function of (X, Y) is

$$p_{X,Y}(x, y) = p_X(x)p_{Y|X}(y|x) = \frac{2}{x^4}\mathbf{1}_{\{x\geq 1,\, 0\leq y\leq x\}}.$$

(c) From the joint density function, we can calculate the marginal density of Y at $y = 1$ as

$$p_Y(1) = \int_{\mathbb{R}} p_{X,Y}(x, 1)\,\mathrm{d}x = \int_1^{\infty} \frac{2}{x^4}\,\mathrm{d}x = \frac{2}{3}.$$

As a result,

$$\mathbb{P}[X > 2 \,|\, Y = 1] = \frac{\int_2^{\infty} p_{X,Y}(x, 1)\,\mathrm{d}x}{p_Y(1)} = \frac{\int_2^{\infty} \frac{2}{x^4}\,dx}{\frac{2}{3}} = \frac{1}{8}.$$

□

Problem 2 (Sleepy Gambler)**.** A gambler is playing a game on a one-dimensional board with positions 0, 1, 2, 3, 4, where

- position 0 means bankruptcy (the gambler loses everything), and
- position 4 means success (the gambler leaves the casino rich).

At each round, if the gambler is currently at position i (with $1 \leq i \leq 3$):

- she moves up to $i + 1$ with probability $\frac{1}{2}(1 - \alpha_i)$,
- she moves down to $i - 1$ with probability $\frac{1}{2}(1 - \alpha_i)$, and
- she stays idle (takes a nap) with probability α_i.

Here we take

$$\alpha_i = \frac{i}{4}$$

to be the "laziness parameter", which depends on her current state i. Intuitively, the closer she is to success, the more time she needs to prepare herself for the next play. Once she reaches 0 or 4, the game ends.

(a) Suppose the gambler starts at position 2. Calculate the probability that the gambler wins within 2 rounds.

(b) Let p_i denote the probability that the gambler eventually succeeds before going bankruptcy, starting from position i. Determine the values of p_i for $i = 1, 2, 3$.

(c) Let μ_i denote the expected number of rounds until the game ends, starting from i. Derive the recursive formula for μ_i and specify boundary conditions. (No need to solve the equations for their values.)

Solution. Let S_n denote the state of the gambler after n rounds.

(1) In order for the gambler to win within 2 rounds, the first two transitions must be $2 \to 3 \to 4$. As a result, the desired probability is

$$\mathbb{P}[S_2 = 4 \,|\, S_0 = 2] = \mathbb{P}[S_2 = 4 \,|S_1 = 3] \cdot \mathbb{P}[S_1 = 3 \,|S_0 = 2] = \frac{1}{8} \cdot \frac{1}{4} = \frac{1}{32}.$$

(2) Let

$$T := \min\{n \geq 0 : \; S_n = 0 \text{ or } 4\}.$$

Using first-step analysis,

$$\begin{aligned} p_i &= \mathbb{P}[S_T = 4 \,|\, S_0 = i] \\ &= \sum_{j \in \{i-1, i, i+1\}} \mathbb{P}[S_T = 4 \,|\, S_1 = j] \cdot \mathbb{P}[S_1 = j \,|\, S_0 = i] \\ &= p_{i-1} \cdot \frac{1}{2}(1 - \alpha_i) + p_i \cdot \alpha_i + p_{i+1} \cdot \frac{1}{2}(1 - \alpha_i). \end{aligned}$$

Re-organizing the equation gives

$$p_i = \frac{1}{2}(p_{i+1} + p_{i-1}), \quad i = 1, 2, 3.$$

This is exactly the same as the recursive equation for the simple symmetric random walk. As a result,

$$p_i = \frac{i}{4}, \quad i = 1, 2, 3.$$

(3) Using first-step analysis,

$$\begin{aligned} \mu_i &= \mathbb{E}[T \,|\, S_0 = i] \\ &= \sum_{j \in \{i-1, i, i+1\}} \mathbb{E}[T \,|\, S_1 = j] \cdot \mathbb{P}[S_1 = j \,|\, S_0 = i] \\ &= (1 + \mu_{i-1}) \cdot \frac{1}{2}(1 - \alpha_i) + (1 + \mu_i) \cdot \alpha_i + (1 + \mu_{i+1}) \cdot \frac{1}{2}(1 - \alpha_i). \end{aligned}$$

Re-organizing the equation gives

$$\mu_i = \frac{1}{1 - \alpha_i} + \frac{1}{2}(\mu_{i+1} + \mu_{i-1}) = \frac{4}{4 - i} + \frac{1}{2}(\mu_{i+1} + \mu_{i-1}), \quad i = 1, 2, 3.$$

The boundary condition is

$$\mu_0 = \mu_4 = 0.$$

□

Problem 3 (Serving Customers)**.** At Tiny World Coffee, customers arrive according to a Poisson process with a rate of two customers every minute. Each customer, independently of anything else, orders a cup of coffee with probability 0.4, orders a cup of non-caffeinated drinks with probability 0.4, and orders a chocolate cookie with probability 0.2. For simplicity, let's assume that each customer orders one and only one item.

(a) On average, how many cups of coffee are sold every minute?

(b) What is the probability that exactly 4 cups of coffee have been sold before the first chocolate cookie is sold? (In this part, the selling of non-caffeinated drinks does not matter.)

(c) Given the information that, by the time the first chocolate cookie is sold, exactly 4 cups of non-caffeinated drinks and 6 cups of coffee have been sold, what is the expected time for the first chocolate cookie to be sold?

Solution. (a) By the principle of Thinning, the rate for the arrivals of coffee buyers is

$$\lambda_{\text{coffee}} = \lambda p_{\text{coffee}} = 2 \times 0.4 = 0.8.$$

(b) By the principle of Superposition and Thinning, if we restrict ourselves to the sequence of customers buying either coffee or cookies, then customers buy coffee with probability $\frac{p_{\text{coffee}}}{p_{\text{coffee}}+p_{\text{cookie}}} = \frac{2}{3}$ and buy cookies with probability $\frac{1}{3}$, independently of each other. So the desired probability is

$$\left(\frac{2}{3}\right)^4 \cdot \frac{1}{3} = \frac{16}{243}.$$

(c) Let T denote the time at which the first cookie is sold, T_{11} denote the arrival time of the 11-th customer, and A denote the event that exactly 4 cups of non-caffeinated drinks and 6 cups of coffee have been sold by time T. We observe that $T = T_{11}$ given A and T_{11} is independent of A by the principle of Superposition and Thinning since A is determined by the types of the arrivals and T_{11} is an arrival time. As a result,

$$\mathbb{E}[T \mid A] = \mathbb{E}[T_{11} \mid A] = \mathbb{E}[T_{11}] = \frac{11}{2},$$

the last equality is due to the fact that T_{11} is Gamma$(11, 2)$-distributed (i.e. it is a sum of 11 independent Exponential(2)-distributed random variables). □